CYBERSECURITY

OVERVIEW

Applying leadership and 21st century skills, participants respond to a cybersecurity challenge by identifying a breach in computer security via "Capture the Flag" games. Areas of challenge might include exploit development, digital puzzles, cryptography, reverse engineering, binary analysis, mobile security, etc. Participants must accurately address a series of on-site problems within a specified, limited amount of time.

ELIGIBILITY

One (1) team of two to four (2-4) members per chapter may participate, one (1) entry per team.

TIME LIMITS

ON-SITE CHALLENGE

A. Teams are required to attend the orientation meeting prior to receiving access to the challenges.

B. Participants have forty-eight (48) hours, beginning at the event orientation meeting, to complete the online challenge.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. Resources and AI Reflection Form referenced in the General Rules is not required for this event.

D. Once registered to compete in this event at the national TSA conference, instructions for registering on the Virginia Tech U.S. Cyber Range Capture The Flag platform will be emailed to chapter advisors.

E. Participants are required to bring their own computer hardware and access to internet.

ON-SITE CHALLENGE

A. Participants report to the event area at the time and place stated in the conference program to attend the orientation meeting and receive the on-site challenge information. TSA competition attire is required to receive the on-site challenge.

B. The event coordinator distributes the materials, information, directions, and deadlines to each team.

C. Teams have forty-eight (48) hours from the designated start time announced during the informational session to complete the online challenge. Teams that do not attend the informational session will not receive additional time and will need to meet with the CRC Manager of the event to participate.

D. Participants will leave the orientation session room once the challenge begins.

E. Solutions are scored in real-time and results will be posted on an online scoreboard. The URL will be provided on-site.

F. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

A. Teams who have not properly registered on the Virginia Tech U.S. Cyber Range Capture the Flag platform prior to conference will not be permitted to participate at the National Conference. Teams will not be permitted to set-up their teams on-site.

B. Participants in this event should concentrate their efforts prior to the competition on researching, understanding, and practicing all aspects of cybersecurity. Please refer to the sample challenge topics listed below.

C. Materials:
   1. Teams are responsible for providing their own computer hardware and any software necessary to solve the on-site challenge.
      a. Participants are required their own computer hardware and access to internet.
      b. Internet access

D. Teams are not to share solutions between teams, included but not limited to online forums, but they may communicate with their own team members. The sharing of information between teams will result in automatic disqualification.

SAMPLE CHALLENGE TOPICS

This list serves only as an example of challenge categories.

A. Web Security
   1. Web Security category often features custom developed web applications which include some web security flaw that must be identified and exploited. Very often SQL injection, command injection, directory traversal, and XSS vulnerabilities are introduced and exploited in these categories.
   2. Examples:
      a. Exploiting poor security controls in a website as a regular user to gain higher level access.
      b. Exploiting poor security practices in a website in order to read arbitrary data from the vulnerable server.
      c. Exploiting a SQL injection vulnerability to extract the content of an intentionally vulnerable server.

B. Forensics
   1. Forensics category often features memory dumps, hidden files, or encrypted data which must be analyzed for information about underlying information.
   2. Examples:
      a. Extracting hidden files from an image of a hard drive.
      b. Extracting hidden files from a memory dump.
      c. Determining the flow of data in a packet capture to ascertain the origin or destination of data.

C. Cryptography
   1. Cryptography is the reason we can use banking apps, transmit sensitive information over the web, and in general protect our privacy. However, a large part of CTFs is breaking widely used encryption schemes that are improperly implemented.
   2. Examples:
      a. Securing web traffic (passwords, communication, etc.).
      b. Securing copyrighted software code.

D. Reverse Engineering
   1. Reverse Engineering category often features programs from all operating systems which must be reverse engineered to determine how the program operates. Typically, the goal is to get the application to reach a certain point or perform some action in order to achieve a solution.
   2. Examples:
      a. Determining what input will cause a program to return True.
      b. Disassembling a game to find a hidden Easter egg not normally accessible or a cheat code to make it easier to win the game.
      c. Optimizing a program to make it run to completion.
      d. Exploiting a buffer overflow with some security mitigations in place to gain a command shell and read a file.
      e. Exploiting a format string vulnerability to gain a command shell and read a file.

EVALUATION

A. Successful completion of the problems is evaluated, as is the time in which it takes teams to complete each challenge.

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Vulnerability Assessor
- Chief Information Security Officer
- Forensic Expert
- Security Architect
- Security Director
- Incident Responder
- Security Manager
- Security Auditor
- Cryptographer
- Security Engineer
- Security Analyst

Participant/Team ID# _____

CYBERSECURITY

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Team is registered on CTF platform

☐ ENTRY NOT EVALUATED

CYBERSECURITY CHALLENGE (55 points)									
1st	**2nd**	**3rd**	**4th**	**5th**	**6th**	**7th**	**8th**	**9th**	**10th**
55 Points	50 Points	45 Points	40 Points	35 Points	30 Points	25 Points	20 Points	15 Points	10 Points
								CYBERSECURITY CHALLENGE SUBTOTAL (55 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL (55 points)**

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

CYBERSECURITY COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator

B. Assistants for set-up and clean-up, two (2) or more.

MATERIALS

A. Coordinator's packet, containing:

   1. Event guidelines, one (1) copy for the coordinator and for each evaluator
   2. TSA Event Coordinator Report

B. Theater seating and a head table for the orientation meeting

C. A copy protocol for the online management materials/ onsite equipment as needed

D. Access to the platform for monitoring and receiving updates and scores of participants.

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.

B. Report to the CRC room and check the contents of the coordinator's packet.

C. Review the event guidelines and check to see that enough evaluators and assistants have been scheduled.

D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

ON-SITE CHALLENGE

A. Begin the event at the scheduled time by closing the doors and checking the entry list.

B. All participants should be in the room at this time. Participants not present will still have the same submission time with no additional time. In order to compete, participants must be on the entry list or must have approval of the CRC.

C. Once teams are seated and general announcements have been given, distribute and review the procedure.

D. After the competition has started, late participants will need to contact the CRC manager to receive the instructions.

E. After the designated time of forty-eight (48) hours has elapsed, the Capture the Flag challenge site will become unavailable.

F. The top ten (10) finalists are announced at the awards ceremony.

G. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.